

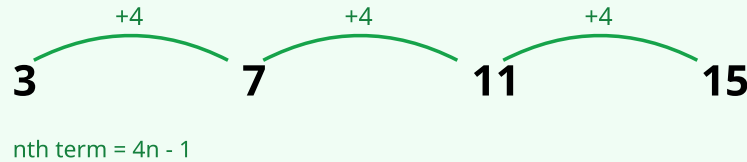
Geometric Sequences

AQA · KS3 Year 7–8

Name: _____ Class: _____ Date: _____

Recognise a geometric sequence, where each term is multiplied by a constant ratio. Find the common ratio, continue the sequence, and find later terms using $\text{term} = a \times r^{(n-1)}$.

VISUAL EXAMPLE



Arithmetic sequence: a constant difference is added each step.

COMMON MISTAKES

Adding a constant instead of multiplying — geometric sequences multiply by the ratio each time.

Finding the ratio the wrong way round — divide a term by the term BEFORE it.

For a decreasing geometric sequence the ratio is a fraction (e.g. $\frac{1}{2}$), not a subtraction.

Using r^n instead of $r^{(n-1)}$ when finding the n th term.

METHOD

- 1 Find the common ratio r by dividing a term by the previous term.
- 2 Check that the same ratio works between every pair of terms.
- 3 To get the next term, multiply the current term by r .
- 4 To find the n th term, use $\text{term} = \text{first term} \times r^{(n-1)}$.
- 5 Check your value against the pattern.

WORKED EXAMPLE

Continue and find a term of: 2, 6, 18, 54, ...

Step 1 — Ratio: $6 \div 2 = 3$, so $r = 3$.

Step 2 — Check: $18 \div 6 = 3$ and $54 \div 18 = 3$

Step 3 — Next term: $54 \times 3 = 162$.

Step 4 — 5th term: $2 \times 3^{(5-1)} = 2 \times 3^4 = 2 \times 81 = 162$.

Step 5 — Matches the next term

1 Write the next two terms of each sequence.

(2 marks)

a) 3, 6, 12, 24, ...

b) 1, 5, 25, 125, ...

HOW TO START

Look at what you multiply by each time, not what you add.

For 3, 6, 12, ..., each term is multiplied by 2, so the next term is $12 \times 2 = 24$.

ANSWERS

(a) = _____

(b) = _____

METHOD 1 Ratio $r = \text{term} \div \text{previous term}$ | 2 Check r is the same throughout | 3 Next term = current $\times r$ | 4 n th term = first $\times r^{(n-1)}$ | 5 Check against the pattern

2 Find the common ratio of each sequence.

(2 marks)

a) 4, 8, 16, ...

b) 2, 6, 18, ...

HOW TO START

Divide a term by the term before it to find the ratio.

For 4, 8, 16: $8 \div 4 = 2$, so $r = 2$.

ANSWERS

(a) = _____

(b) = _____

METHOD 1 Ratio $r = \text{term} \div \text{previous term}$ | 2 Check r is the same throughout | 3 Next term = current $\times r$ | 4 n th term = first $\times r^{(n-1)}$ | 5 Check against the pattern

3 Write the next two terms.

(3 marks)

a) 5, 15, 45, ...

b) 80, 40, 20, ...

c) 1, -2, 4, -8, ...

ANSWERS

(a) = _____

(b) = _____

(c) = _____

METHOD 1 Ratio $r = \text{term} \div \text{previous term}$ | 2 Check r is the same throughout | 3 Next term = current $\times r$ | 4 n th term = first $\times r^{(n-1)}$ | 5 Check against the pattern

4 Find the term described.

(3 marks)

- a) First term 3, ratio 2 — find the 4th term.
- b) First term 2, ratio 5 — find the 3rd term.
- c) First term 100, ratio $\frac{1}{2}$ — find the 4th term.

ANSWERS

- (a)** = _____
- (b)** = _____
- (c)** = _____

METHOD 1 Ratio $r = \text{term} \div \text{previous term}$ | 2 Check r is the same throughout | 3 Next term = current $\times r$ | 4 nth term = first $\times r^{(n-1)}$ | 5 Check against the pattern

5 State whether each sequence is arithmetic or geometric, and give the common difference or ratio. (3 marks)

a) 2, 4, 6, 8, ...

b) 2, 4, 8, 16, ...

c) 5, 10, 20, 40, ...

ANSWERS

(a) = _____

(b) = _____

(c) = _____

METHOD 1 Ratio $r = \text{term} \div \text{previous term}$ | 2 Check r is the same throughout | 3 Next term = current $\times r$ | 4 n th term = first $\times r^{(n-1)}$ | 5 Check against the pattern

6 A colony of bacteria doubles in number every hour. It starts with 50 bacteria. (3 marks)

a) Write the number of bacteria after 0, 1, 2 and 3 hours. (2 marks)

b) How many bacteria are there after 5 hours? (1 mark)

ANSWERS

(a) = _____

(b) = _____

METHOD 1 Ratio $r = \text{term} \div \text{previous term}$ | 2 Check r is the same throughout | 3 Next term = current $\times r$ | 4 n th term = first $\times r^{(n-1)}$ | 5 Check against the pattern

A geometric sequence has first term 4 and common ratio 3.

(4 marks)

- Write the first four terms. (1 mark)
- Find the 6th term. (1 mark)
- Which term has the value 2916? (2 marks)

ANSWERS

(a) = _____

(b) = _____

(c) = _____

METHOD 1 Ratio $r = \text{term} \div \text{previous term}$ | 2 Check r is the same throughout | 3 Next term = current $\times r$ | 4 n th term = first $\times r^{(n-1)}$ | 5 Check against the pattern

8 A ball is dropped from 20 m. After each bounce it reaches 0.8 of its previous height. (5 marks)

- a) Find the height after the 1st, 2nd and 3rd bounce. (2 marks)
b) Is the sequence of bounce heights arithmetic or geometric? Explain. (1 mark)
c) Will the bounce height ever reach exactly 0? Explain your answer. (2 marks)

ANSWERS

- (a) = _____
(b) = _____
(c) = _____

METHOD 1 Ratio $r = \text{term} \div \text{previous term}$ | 2 Check r is the same throughout | 3 Next term = current $\times r$ | 4 nth term = first $\times r^{(n-1)}$ | 5 Check against the pattern

Self-reflection

How confident do you feel with this topic now? (tick one)

Not yet

Getting there

Confident

Really confident

Which question did you find the hardest, and why?

How can I tell a geometric sequence from an arithmetic one?

One thing I will practise before the next lesson:

Teacher key

Q1: a) 48, 96 ($\times 2$) b) 625, 3125 ($\times 5$)

Q2: a) $r = 2$ b) $r = 3$

Q3: a) 135, 405 ($\times 3$) b) 10, 5 ($\times \frac{1}{2}$) c) 16, -32 ($\times -2$)

Q4: a) $3 \times 2^3 = 24$ b) $2 \times 5^2 = 50$ c) $100 \times (\frac{1}{2})^3 = 12.5$

Q5: a) arithmetic, $d = 2$ b) geometric, $r = 2$ c) geometric, $r = 2$

Q6: a) 50, 100, 200, 400 b) $50 \times 2^5 = 1600$ bacteria

Q7: a) 4, 12, 36, 108 b) $4 \times 3^5 = 972$ c) $4 \times 3^{(n-1)} = 2916$ $3^{(n-1)} = 729 = 3^6$ $n - 1 = 6$ $n = 7$ (the 7th term)

Q8: a) $20 \times 0.8 = 16$; $16 \times 0.8 = 12.8$; $12.8 \times 0.8 = 10.24$ m b) geometric, each height is $\times 0.8$ c) No — multiplying by 0.8 makes the height smaller each time but never exactly 0, so it gets closer and closer to 0 without reaching it
